

Unnamed_Unit

unit 5 Knowledge Check STEAM Questions & Critical Thinking

PDF Documents

PDF Document 1: 1731845260047-Chapter 6_ unit5 Knowledge Check.pdf

This PDF document is included in the unit materials.

KNOWLEDGE CHECK

1 What color of fire is typically the hottest?

- ☐ A Red
- ☐ B Orange
- ☐ C Blue
- ☐ D White

CORRECT ANSWER

☐ D

2 Which LED color did we program to represent the lowest temperature?

- ☐ A Red
- ☐ B Orange
- ☐ C Blue
- ☐ D White

CORRECT ANSWER

☐ A

3 What happens if you do not use a wait block in your code?

- ☐ A The LED will not light up
- ☐ B The changes between colors will be too fast to see
- ☐ C The code will stop running
- ☐ D There will be no effect

CORRECT ANSWER

☐ B

4 What does a variable in programming do?

- ☐ A Tells the computer to turn off
- ☐ B Stores a piece of information that can change
- ☐ C Makes the LED lights blink
- ☐ D Connects to the internet

CORRECT ANSWER

☐ C

5 If the variable value is set to 3 in the 3-brightness lamp , which color will the LED display in our mood lamp project?

- ☐ A Red
- ☐ B Orange
- ☐ C Blue
- ☐ D White

CORRECT ANSWER

☐ C

6 Which module did we use to control the changes in the LED colors?

- ☐ A Network Module
- ☐ B Button Module
- ☐ C Display Module
- ☐ D Speaker Module

CORRECT ANSWER

☐ B

7 True or False: Orange flames are typically cooler than blue flames.

- ☐ A True
- ☐ B False

CORRECT ANSWER

☐ A

8 Why is it important to understand the temperature and color relationship in fires?

- ☐ A It helps in cooking
- ☐ B It is only important for scientists
- ☐ C It can help in determining the safety of a fire
- ☐ D It affects the speed of the internet

CORRECT ANSWER

☐ C

 Critical Thinking Challenge

- 1 How does understanding the color and temperature of fire help us in everyday life? Provide a brief example.

- 2 What does 'variable' mean in programming? Can you think of something similar in everyday life? Explain how it works.